

## F. The Spirit-Oak's Resonance

time limit per test: 1.5 seconds  
memory limit per test: 256 megabytes

After narrowly escaping the Beast's game, the three friends found themselves in a vast, ancient grove dominated by colossal, glowing Spirit-Oaks. These trees, according to ABD-ELMOHAYMEN's ever-present "Unusual Flora of the Unknown" guide, resonate with different frequencies of spirit energy, measured by unique 'Resonance Values'. The forest was divided into  $N$  distinct segments, and each segment  $i$  had a Spirit-Oak with a specific resonance value  $A_i$ .

Their immediate goal was to find a 'Resonance Core' – a mystical artifact that required a precise combination of spirit energies to activate. The guide indicated that the Core would only appear if they could accurately determine the total 'strength' of a particular range of spirit energies within a specific section of the forest.

Periodically, a query would appear on a shimmering leaf: "How many Spirit-Oaks within segments  $L$  to  $R$  (inclusive) possess a Resonance Value between  $X$  and  $Y$  (inclusive)?"

OMAR quickly realized this wasn't just about counting individual trees, but about summing up the presence of all trees whose resonance fell within a specific range. They knew the forest segments stretched for miles, and there would be many such queries. Only an efficient method would allow them to progress.

Can you help them calculate, for each query, the total count of Spirit-Oaks whose Resonance Values fall within the specified range  $[X, Y]$  in the given forest segment  $[L, R]$  ?

### Input

The first line contains two integers  $N(1 \leq N \leq 10^5)$ , the number of forest segments, and  $Q(1 \leq Q \leq 10^5)$ , the number of queries.

The second line contains  $N$  integers  $A_1, A_2, \dots, A_N (1 \leq A_i \leq 10^5)$ , representing the Resonance Value of the Spirit-Oak in each segment.

The next  $Q$  lines each contain four integers  $L, R, X, Y(1 \leq L \leq R \leq N, 1 \leq X \leq Y \leq 10^5)$ , representing the start and end of the forest segment, and the lower and upper bounds for the Resonance Values.

### Output

For each query, print a single integer on a new line, representing the total count of Spirit-Oaks meeting the criteria.

### Example

| input                                                                        | Copy |
|------------------------------------------------------------------------------|------|
| 7 5<br>1 2 1 3 2 7 2<br>2 5 1 2<br>1 4 1 1<br>1 6 2 3<br>7 7 2 2<br>1 7 8 10 |      |
| output                                                                       | Copy |
| 3<br>2<br>3<br>1<br>0                                                        |      |

**Game of Coders 4 - Over the Garden Wall**

Finished

Practice

→ About Contest



Contest website

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)▼

Choose file: Choose File No file chosen

Submit

→ Last submissions

| Submission                | Time              | Verdict  |
|---------------------------|-------------------|----------|
| <a href="#">332714099</a> | Aug/07/2025 03:07 | Accepted |
| <a href="#">331018074</a> | Jul/27/2025 00:00 | Accepted |

→ Contest materials

Statements (pdf) (en)